PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

This paper was submitted to a another journal from BMJ but declined for publication following peer review. The authors addressed the reviewers' comments and submitted the revised paper to BMJ Open. The paper was subsequently accepted for publication at BMJ Open.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Values, Preferences and Burden of Treatment for the Initiation of GLP-1 Receptor Agonists and SGLT-2 Inhibitors in Adult Patients with Type 2 Diabetes: A Systematic Review
AUTHORS	González-González, José Gerardo; Díaz González-Colmenero, Alejandro; Millán-Alanís, Juan Manuel; Lytvyn, Lyubov; Solis, Ricardo Cesar; Mustafa, Reem; Palmer, Suetonia; Li, Sheyu; Hao, Qiukui; Alvarez-Villalobos, Neri; Vandvik, Per; Rodríguez-Gutiérrez, René

VERSION 1 – REVIEW

REVIEWER	Lotte Klim Studies&Me I have included my name in my comments for the author
REVIEW RETURNED	10-Dec-2020

GENERAL COMMENTS	Dear Editor and Authors Thank you for the opportunity to review this manuscript. Overall the manuscript is very clear and describes the reveiw-process and decision points very well. However, I am left with only one questions that could add value to how results may be used. It would be highly informative to know the level of validity of the questionnaires that were accepted for the analysis. It could be in terms of patient input for which items to include, psychometric methods, acceptablilty etc. - or any considerations on the consequences of lack of this information.

REVIEWER	Kristina Boye Eli Lilly and Company I have included my name in my comments for the author
REVIEW RETURNED	31-Dec-2020

GENERAL COMMENTS	- Thank you for the opportunity to review this interesting article. - In the discussion, it states "However, these results demonstrate a major shortcoming of our systematic review; none of the studies presented patients with best current evidence on benefits and harms of these drugs, making any inferences about values and preferences of highly limited value." This seems to the major issue with the research work found on this topic. However, the authors fail to address or realize a vital/practical point: Where would this 'current evidence on benefit and harms of

	these drugs' come from specifically? One would think it would come from published clinical trials but unfortunately there are not many head to head clinical trials that are available to directly allow for the comparisons to be considered in research. If on the other hand, meta-analyses should be conducted to compare preferences between agents, then that causes people to not only investigate the preference study itself but also what is and what is not included in -or not included in- the meta-analysis, if a direct H2H clinical trail is not available. Thus, would the authors view the research differently based on the availability of head to head clinical trial data being available or not? This was likely a common reason why some preference studies either did or did not include a wider variety of attributes or not. - It would be helpful if the authors would provide some thoughts on why no preference studies for SGLT-2 agents could be found. - The authors end with the sentence: "Our review findings emphasize the importance of standardizing evaluation instruments in order to ensure precise and trustworthy results." May the authors explain exactly what type of standard instruments they would like used? Indeed, standard methods were used in a vast majority of the studies (TTO, DCE) but how would a standard instrument be developed if there are different attributes (e.g., side effects, devices) that could occur with different and newer therapies? For example, there are some standard quality of life questionnaires developed in the diabetes field but as new agents are created -for example those who now help with CV health or have different injection or oral therapy characteristics- should these older out of date standard instruments be used just because they are standardized?

REVIEWER	Jeremy Sussman University of Michigan I have included my name in my comments for the author
REVIEW RETURNED	02-Jan-2021

GENERAL COMMENTS	This is a systematic review of values and preferences for SGLT2 and GLP-1 medications as part of the BMJ living review series. This is a valuable series and the mechanism of the review is well done. They unfortunately didn't find any studies of SGLT2, which is surprising and disappointing, but also an interesting finding. They found a reasonable number of GLP-1 studies. I have no major shifts, but did think there are moderately large number of narrative changes that would improve readability. The structure of the paper, by having large tables with no figures or other summarizing results, makes it quite hard to see the big picture. I agree with the decision not to attempt forest plots, which would compare apples and oranges and be misleading. I don't see an easy way to make the tables clearer for summarization, though it would be nice. In the results, in general, the formulation "[Author] found" is not very interesting or informative. Re-writing the results to focus on the questions and their findings would read better. So, for example, I would change a sentence like "Polster et al found" to "an online survey with a conjoint analysis." This happens roughly 5 times.

	However, this does make it even more important to be clear about which findings are meaningful, large, or consistent. Results paragraphs also would be clearer if take-home messages were early in the paragraph, but right now are often at the end. This reads well: “Two studies evaluated preferences between … both of them showed preference for GLP-q RA.” But “Attributes that were included in all studies” would make more sense earlier in the paragraph. The first paragraph of the discussion, which is unusually important in this paper for the reasons above, should be made more declarative. The sentence “notable findings from these studies” is passive voice, hard to read, and I struggled to know what the finding is. Changing it to “Patients consistently showed resistance to injectables and complicated devices, preferring oral medications.” The phrase “none of the studies presented patients with current evidence” doesn’t make clear how, why not, or how current evidence is different from what the studies did use. The rest of that paragraph has this issue also. There is a small definitional issue in table 1. The methodology column conflates the assessment approach (TTO, DCE) with the study design (observational, crossover). But one could easily have a crossover TTO, etc. I think I would recommend calling it something like “assessment approach” and using TTO, DCE, ranking, and rating. For Figure 2 please format so you can see the questions. It’s so much harder that way. See, for example, the equivalent fig 2 of BMJ 2016;355:i6112.

REVIEWER	Margrethe Nielsen Copenhagen University College I have included my name in my comments for the author
REVIEW RETURNED	04-Jan-2021

GENERAL COMMENTS	It is a relevant and interesting research question and even if the study is unable to reach a conclusion to the research question the findings are still relevant and pointing at the need for further research to be able to inform both patients and health professionals. I have a few comments to the study. 1. It is mentioned in the section Eligibility criteria that any study using quantitative or quality analysis are included. As far as I can judge from table 1 the studies included are quantitative? I recommend that it is commented on which studies are qualitative and if any, how they are evaluated with respect to risk of bias and evidence. 2. I miss arguments for the exclusion of the mentioned study designs. 3. In the section Search strategy there is a reference to a appendix 1 which is not included. I think the relevant document is called supplementary file 1: examples of search strategy. 4. I agree with the authors who states that the study has multiple strengths and also limitations, but find that it is important to summarize how much or little is known which can inform both patients and health professionals when choice of treatment is to be taken. 5. I recommend that the study is published.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Comments:
Dear Editor and Authors
Thank you for the opportunity to review this manuscript.
Overall the manuscript is very clear and describes the review-process and decision points very well. However, I am left with only one questions that could add value to how results may be used. It would be highly informative to know the level of validity of the questionnaires that were accepted for the analysis. It could be in terms of patient input for which items to include, psychometric methods, acceptablilty etc. - or any considerations on the consequences of lack of this information.

Answer: Thank you for your comment, we in fact agree with your point. We have discussed the impact of the validity of questionnaires on probably biasing results across studies. As part of our risk of bias assessment, we also analyzed the validity of each employed preference assessment instrument across studies so that this point was taken into consideration.

Reviewer: 2

Comments:
- Thank you for the opportunity to review this interesting article.
- In the discussion, it states "However, these results demonstrate a major shortcoming of our systematic review; none of the studies presented patients with best current evidence on benefits and harms of these drugs, making any inferences about values and preferences of highly limited value." This seems to the major issue with the research work found on this topic. However, the authors fail to address or realize a vital/practical point: Where would this 'current evidence on benefit and harms of these drugs' come from specifically? One would think it would come from published clinical trials but unfortunately there are not many head to head clinical trials that are available to directly allow for the comparisons to be considered in research. If on the other hand, meta-analyses should be conducted to compare preferences between agents, then that causes people to not only investigate the preference study itself but also what is and what is not included in -or not included in- the meta-analysis, if a direct H2H clinical trail is not available. Thus, would the authors view the research differently based on the availability of head to head clinical trial data being available or not? This was likely a common reason why some preference studies either did or did not include a wider variety of attributes or not.

Answer: Thank you for your kind comments. Regarding this topic, we do not necessarily refer to head-to-head trials being performed. We rather mean that especially in the field of diabetes treatment evidence changes very quickly and that means treatment profiles also tend to change as well, therefore this could be a limitation for studies evaluating preferences for different treatments. We have tried to make that point more clearer in the discussion section.

- It would be helpful if the authors would provide some thoughts on why no preference studies for SGLT-2 agents could be found.

Answer: We have added some thoughts on the matter in the discussion section.

- The authors end with the sentence: "Our review findings emphasize the importance of standardizing evaluation instruments in order to ensure precise and trustworthy results." May the authors explain exactly what type of standard instruments they would like used? Indeed, standard methods were used in a vast majority of the studies (TTO, DCE) but how

would a standard instrument be developed if there are different attributes (e.g., side effects, devices) that could occur with different and newer therapies? For example, there are some standard quality of life questionnaires developed in the diabetes field but as new agents are created -for example those who now help with CV health or have different injection or oral therapy characteristics- should these older out of date standard instruments be used just because they are standardized?

Answer: In this case, we were not referring to the study methodology or assessment method but rather to the attributes that we consider should always be mentioned in a treatment profile. If not standardized, incomplete treatment profiles could be presented to patients and bias their preferences based on uninformed decisions. We have made some shifts in the discussion section to make this section clearer.

Reviewer: 3

Recommendation:

Comments:
This is a systematic review of values and preferences for SGLT2 and GLP-1 medications as part of the BMJ living review series. This is a valuable series and the mechanism of the review is well done. They unfortunately didn't find any studies of SGLT2, which is surprising and disappointing, but also an interesting finding. They found a reasonable number of GLP-1 studies.

I have no major shifts, but did think there are moderately large number of narrative changes that would improve readability.

The structure of the paper, by having large tables with no figures or other summarizing results, makes it quite hard to see the big picture. I agree with the decision not to attempt forest plots, which would compare apples and oranges and be misleading. I don't see an easy way to make the tables clearer for summarization, though it would be nice.

In the results, in general, the formulation "[Author] found" is not very interesting or informative. Re-writing the results to focus on the questions and their findings would read better. So, for example, I would change a sentence like "Polster et al found" to "an online survey with a conjoint analysis." This happens roughly 5 times.

Answer: Thank you for your useful comment, we have performed the corresponding changes in the results section.

However, this does make it even more important to be clear about which findings are meaningful, large, or consistent.

Results paragraphs also would be clearer if take-home messages were early in the paragraph, but right now are often at the end. This reads well: "Two studies evaluated preferences between … both of them showed preference for GLP-q RA." But "Attributes that were included in all studies" would make more sense earlier in the paragraph.

Answer: We have performed the corresponding changes in the results section and placed the most important messages at the beginning of each section to facilitate comprehension of results.

The first paragraph of the discussion, which is unusually important in this paper for the reasons above, should be made more declarative. The sentence "notable findings from these studies" is

passive voice, hard to read, and I struggled to know what the finding is. Changing it to “Patients consistently showed resistance to injectables and complicated devices, preferring oral medications.” The phrase “none of the studies presented patients with current evidence” doesn’t make clear how, why not, or how current evidence is different from what the studies did use. The rest of that paragraph has this issue also.

Answer: We have followed your advice on being more declarative and providing more insight in the last part of the first paragraph of the discussion section.

There is a small definitional issue in table 1. The methodology column conflates the assessment approach (TTO, DCE) with the study design (observational, crossover). But one could easily have a crossover TTO, etc. I think I would recommend calling it something like “assessment approach” and using TTO, DCE, ranking, and rating.

Answer: We have changed it to “assessment approach”, thank you for the observation.

For Figure 2 please format so you can see the questions. It’s so much harder that way. See, for example, the equivalent fig 2 of BMJ 2016;355:i6112

Answer: Thank you for the observation, we have included the name of each domain in the figure 2 in order to make it easier to comprehend.

Reviewer: 4

Comments:
It is a relevant and interesting research question and even if the study is unable to reach a conclusion to the research question the findings are still relevant and pointing at the need for further research to be able to inform both patients and health professionals.
I have a few comments to the study.
1. It is mentioned in the section Eligibility criteria that any study using quantitative or quality analysis are included. As far as I can judge from table 1 the studies included are quantitative? I recommend that it is commented on which studies are qualitative and if any, how they are evaluated with respect to risk of bias and evidence.

Answer: Thank you for the observation, we did not find any qualitative study and have added this statement in the results section.

2. I miss arguments for the exclusion of the mentioned study designs.

Answer: We have added the arguments for excluding the mentioned study designs in the methods section.

3. In the section Search strategy there is a reference to a appendix 1 which is not included. I think the relevant document is called supplementary file 1: examples of search strategy.

Answer: We have made the corresponding change in the manuscript.

4. I agree with the authors who states that the study has multiple strengths and also limitations, but find that it is important to summarize how much or little is known which can inform both patients and health professionals when choice of treatment is to be taken.

Answer: It is indeed an important topic to mention, we have added it in the discussion section in order to simplify a take home message for the reader. Thank you for the observation.

5. I recommend that the study is published.